ESCOLA
SUPERIOR
DE MEDIA
ARTES
E DESIGN
POLITÉCNICO
DO PORTO

ALP

FICHA DE EXERCÍCIOS
ATIVIDADE LETIVA

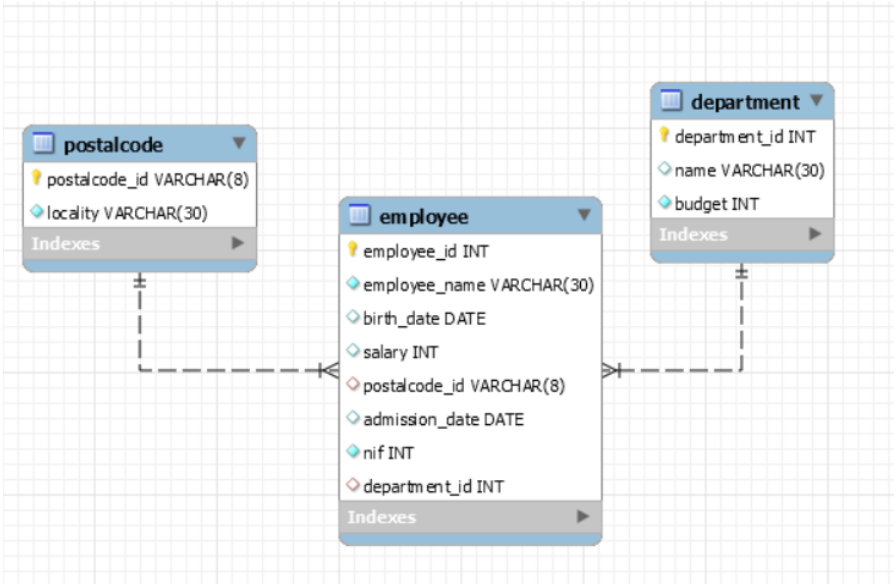
Databases

UNIDADE CURRICULAR

SQL Language

FICHA

Consider the ***Company*** database, with the following scheme:



Construct the SQL Queries to obtain the following results:

1. Name of department with larger budget

Result Grid

name
Computing

2. Name of department with lower budget

Result Grid

name
Sales

3. List of employees of department “Production”

employee_id	employee_name	birth_date	salary	postalcode_id	admission_date	nif	department_id
1	Manuel Santos	1972-01-01	1550	4000-100	2007-03-03	123456789	1
2	Paulo Fosneca	1973-03-03	1820	4000-100	2008-01-01	234567890	1
3	Carla Carolina	1974-04-04	1550	4100-050	2008-01-01	345678901	1
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. List of employees of departments "*Production*", "*Accounting*" and "*Sales*"

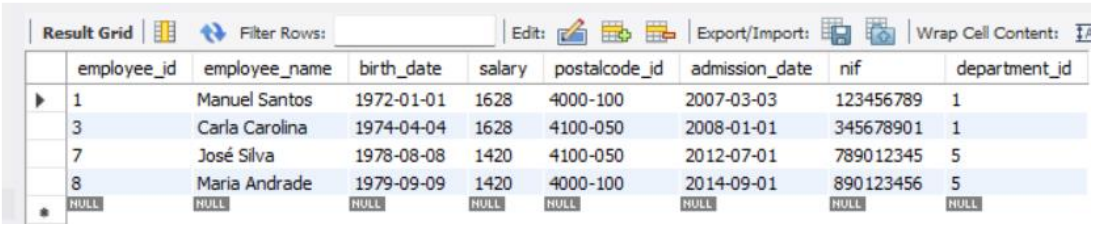


employee_id	employee_name	birth_date	salary	postalcode_id	admission_date	nif	department_id
1	Manuel Santos	1972-01-01	1628	4000-100	2007-03-03	123456789	1
2	Paulo Fosneca	1973-03-03	1911	4000-100	2008-01-01	234567890	1
3	Carla Carolina	1974-04-04	1628	4100-050	2008-01-01	345678901	1
5	Maria Costa	1976-06-06	1950	4480-876	2010-01-01	567890123	2
6	Ricardo Rocha	1977-07-07	2150	4480-876	2012-07-01	678901234	4
9	Liliana Lousada	1980-10-10	2350	4460-100	2014-09-01	901234567	4

5. Number of employees by department

COUNT(*)	name
3	Production
1	Accounting
2	Computing
2	Sales
2	Logistics

6. List employees with salary below average salary



employee_id	employee_name	birth_date	salary	postalcode_id	admission_date	nif	department_id
1	Manuel Santos	1972-01-01	1628	4000-100	2007-03-03	123456789	1
3	Carla Carolina	1974-04-04	1628	4100-050	2008-01-01	345678901	1
7	José Silva	1978-08-08	1420	4100-050	2012-07-01	789012345	5
8	Maria Andrade	1979-09-09	1420	4000-100	2014-09-01	890123456	5

7. insert employee below, through query

employee_id	name	birth_date	salary	postalcode_id	admission_date	nif	department_id
7	José Silva	1978-08-08	1420	4100-050	2012-07-01	789012345	5
8	Maria Andrade	1979-09-09	1420	4000-100	2014-09-01	890123456	5
9	Liliana Lousada	1980-10-10	2350	4460-100	2014-09-01	901234567	4
10	Diogo Dionísio	1981-11-11	2205	4460-100	2014-09-01	213456789	3
11	Verónica Santos	1998-02-03	2100	4000-100	2018-02-01	345678901	5

8. Insert, through query, a new record on table department

department_id	name	budget
1	Produção	12500
2	Contabilidade	4780
3	Projetos	7000
4	Recursos Humanos	5200
5	Logística	3950
6	teste	5000

9. List departments that do not have employees

department_id	name	budget
6	teste	5000

10. Edit salary of employees who belong to the *Production* department, increasing them by 5%

11. Remove the department that you have created earlier (department 6, by SQL statement)

12. Edit salary of employees admitted in 2008, increasing them by 50€